§3.4.

P1. Solution: $\dot{X} = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix} X$

$$\Rightarrow \frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = -x_1 - x_2$$

$$\Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 0 \quad \text{by letting } x_1 = y,\ x_2 = \frac{dy}{dt}$$

$$r^2 + r + 1 = 0$$

$$r = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$\Rightarrow y_1(t) = e^{-\frac{t}{2}} \cos\frac{\sqrt{3}}{2}t,\quad y_2(t) = e^{-\frac{t}{2}} \sin\frac{\sqrt{3}}{2}t$$

$$\Rightarrow X^1(t) = \begin{pmatrix} y_1(t) \\ y_1'(t) \end{pmatrix} = \begin{bmatrix} e^{-\frac{t}{2}} \cos\frac{\sqrt{3}}{2}t \\ e^{-\frac{t}{2}}\left(-\frac{1}{2}\cos\frac{\sqrt{3}}{2}t - \frac{\sqrt{3}}{2}\sin\frac{\sqrt{3}}{2}t\right) \end{bmatrix}$$

$$X^2(t) = \begin{pmatrix} y_2(t) \\ y_2'(t) \end{pmatrix} = \begin{bmatrix} e^{-\frac{t}{2}} \sin\frac{\sqrt{3}}{2}t \\ e^{-\frac{t}{2}}\left(-\frac{1}{2}\sin\frac{\sqrt{3}}{2}t + \frac{\sqrt{3}}{2}\cos\frac{\sqrt{3}}{2}t\right) \end{bmatrix}$$

are a basis for the given differential equation.

P2. Solution: $\dot{X} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -1 & 2 \end{pmatrix} X$

$$\Rightarrow \frac{dx_1}{dt} = x_2$$

$$\frac{dx_2}{dt} = x_3$$

$$\frac{dx_3}{dt} = 2x_1 - x_2 + 2x_3$$

$$\Rightarrow \frac{d^3y}{dt^3} - 2\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = 0$$

by letting $x_1 = y$, $x_2 = \frac{dy}{dt}$, $x_3 = \frac{d^2y}{dt^2}$

$$r^3 - 2r^2 + r - 2 = 0$$

$$r^2(r-2) + r - 2 = 0$$

$$(r-2)(r^2+1) = 0$$

$$\Rightarrow r_1 = 2,\ r_2 = i,\ r_3 = -i$$

$$\Rightarrow y_1(t) = e^{2t},$$

$$y_2(t) = \cos t,\ y_3(t) = \sin t.$$

$$\Rightarrow X^1(t) = \begin{bmatrix} y_1(t) \\ y_1'(t) \\ y_1''(t) \end{bmatrix} = \begin{bmatrix} e^{2t} \\ 2e^{2t} \\ 4e^{2t} \end{bmatrix}$$

$$X^2(t) = \begin{bmatrix} y_2(t) \\ y_2'(t) \\ y_2''(t) \end{bmatrix} = \begin{bmatrix} \cos t \\ -\sin t \\ -\cos t \end{bmatrix}$$

$$X^3(t) = \begin{bmatrix} y_3(t) \\ y_3'(t) \\ y_3''(t) \end{bmatrix} = \begin{bmatrix} \sin t \\ \cos t \\ -\sin t \end{bmatrix}$$

are a basis for the given differential equation.

P5. Solution: $\dot{X} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} X$

$$X^1(t) = \begin{pmatrix} e^t \\ -e^t \end{pmatrix}$$

$$\dot{X}^1(t) = \begin{pmatrix} e^t \\ -e^t \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} e^t \\ -e^t \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} X^1(t) \Rightarrow X^1(t) \text{ is a solution}$$

$$X^2(t) = \begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$$

$$\dot{X}^2(t) = \begin{pmatrix} -e^{-t} \\ -e^{-t} \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} e^{-t} \\ e^{-t} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} X^2(t) \Rightarrow X^2(t) \text{ is a solution.}$$

$$X^1(0) = \begin{pmatrix} 1 \\ -1 \end{pmatrix},\quad X^2(0) = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

It is easy to see $X^1(0)$ and $X^2(0)$ are linearly independent, so that $X^1(t)$ and $X^2(t)$ are linearly independent. Thus, $X^1(t)$ and $X^2(t)$ form a basis for the set of all solutions. Yes.

P6. Solution: $\dot{X} = \begin{pmatrix} 4 & -2 & 2 \\ -1 & 3 & 1 \\ 1 & -1 & 5 \end{pmatrix} X$

$$X^1(t) = \begin{pmatrix} e^{2t} \\ e^{2t} \\ 0 \end{pmatrix} \Rightarrow X^1(0) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

$$X^2(t) = \begin{pmatrix} 0 \\ e^{4t} \\ e^{4t} \end{pmatrix} \Rightarrow X^2(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$X^3(t) = \begin{pmatrix} e^{6t} \\ 0 \\ e^{6t} \end{pmatrix} \Rightarrow X^3(0) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$c_1 X^1(0) + c_2 X^2(0) + c_3 X^3(0) = 0$

$$c_1\begin{pmatrix}1\\1\\0\end{pmatrix} + c_2\begin{pmatrix}0\\1\\1\end{pmatrix} + c_3\begin{pmatrix}1\\0\\1\end{pmatrix} = 0$$

$$\left.\begin{array}{l} c_1 + c_3 = 0 \\ c_1 + c_2 = 0 \end{array}\right\} \Rightarrow c_2 = c_3$$

$$c_2 + c_3 = 0 \quad \Rightarrow c_2 = c_3 = 0 \Rightarrow c_1 = 0$$

$\Rightarrow X^1(0), X^2(0), X^3(0)$ are linearly independant.

$\Rightarrow X^1(t), X^2(t), X^3(t)$ are linearly independant solutions

$\Rightarrow X^1(t), X^2(t), X^3(t)$ form a basis for the set of all solutions. Yes

P8. Solution: The dimension of the set of all solutions of $\dot{X} = \begin{pmatrix}-5 & 2 & -2\\ 1 & -4 & -1\\ -1 & 1 & -6\end{pmatrix} X$ is 3. But here we only have two vectors $X^1(t) = \begin{bmatrix}e^{-3t}\\ e^{-3t}\\ 0\end{bmatrix}$ and $X^2(t) = \begin{bmatrix}0\\ e^{-5t}\\ e^{-5t}\end{bmatrix}$. It is impossible for $X^1(t)$ and $X^2(t)$ form a basis for the set of all solutions. No